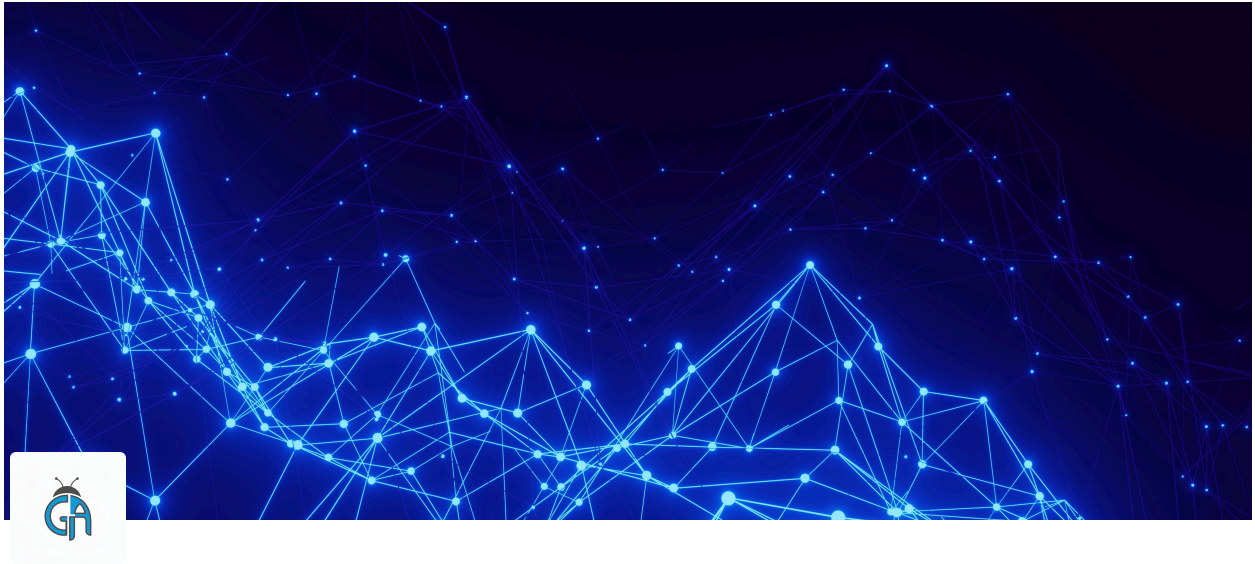




Static vs. Dynamic Testing

Reading vs. Running

 Tags: QA, Fundamentals, Interview Prep

 Reading Time: 3 min

TL;DR

Static Testing checks the documents and code *without* running the software (Prevention).

Dynamic Testing checks the functionality *while* running the software (Cure).

The Core Difference

The difference is simple: Is the code running?

Static Testing (The Verification)

- "Verification: Are we building the product right?"
- We examine the **documentation** (requirements, designs) or the **code structure**.
- **Goal:** Find errors early (typos, logic gaps, missing requirements).
- **Techniques:** Reviews, Walkthroughs, Inspections, Proofreading.

Dynamic Testing (The Validation)

- "Validation: Are we building the right product?"
 - We execute the **software**. We input data and check the output.
 - **Goal:** Find bugs that happen during execution (crashes, wrong calculations).
 - **Techniques:** Unit Testing, Integration Testing, System Testing (Manual execution).
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The Car Analogy

Static Testing (The Blueprint Check):

You look at the blueprints for the car engine. You notice the fuel line is drawn too close to the exhaust pipe.

You found a disaster before the car was even built. The engine is OFF.

Dynamic Testing (The Test Drive):

You build the car, turn on the engine, and drive it. You hear a rattling noise when you hit 60mph.

You found a bug while using the product. The engine is ON.

The Comparison Matrix

Component: State of Code

- Static: Code is NOT executed.
- Dynamic: Code IS executed.

Component: What we find

- Static: Logic errors, missing requirements, syntax errors, bad design.
- Dynamic: Functional failures, performance issues, memory leaks.

Component: Cost to Fix

- Static: 💰 Low (Changing a document is cheap).
- Dynamic: 💰💰💰 High (Rewriting code and re-testing is expensive).

Component: Who?

- Static: Business Analysts, Developers, QAs (Reviewing stories).
 - Dynamic: QAs (Testing the app), Developers (Unit tests).
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💡 Real World Example

Scenario: A Login Feature

🛑 Static Test (Reviewing the Requirement):

The requirement says: *"User password must be 8 characters."*

You ask: *"What about special characters? What about numbers? This requirement is incomplete."*

Result: You prevented a bug just by reading.

🏃 Dynamic Test (Running the App):

The app is built. You try to create a password with 7 characters.

Result: The app accepts it (Bug!). You report it.

🧠 Why This Matters

"Static Testing" is why I ask questions during the **Requirement Analysis** phase.

Finding a bug in the documentation costs \ \$1 to fix. Finding that same bug after it is coded costs \ \$100 to fix.